



A quick-reference guide for tcpdump packet capture: command-line options, BPF filter syntax, and TCP/IP header layout for traffic analysis.

tcpdump [-aAenStvxX] [-F filterfile] [-i interface] [-c count] [-r pcapfile] [-s snaplen] [-w pcapfile] ['bpf filter']

tcpdump Usage

Useful One-Liners

ICMP Types

OPTION	MEANING
-A	Display each packet's payload in ASCII.
-c n	Exit after capturing n packets.
-D	List available network interfaces.
-e	Print the data link layer (Ethernet) header.
-F file	Read the BPF filter expression from a file.
-i iface	Listen on the specified interface.
-n	Do not resolve IP addresses or port names.
-r file	Read packets from a capture file.
-s snaplen	Set the snapshot length in bytes.
-S	Print absolute TCP sequence numbers.
-t	Do not print a timestamp on each line.
-tttt	Print a date-and-time timestamp.
-v / -vv / -vvv	Increase verbosity (more v = more detail).
-w file	Write raw packets to a pcap file.
-X	Print each packet in hex.
-xx	Print the link layer header in hex too.
-X	Print payload in hex <i>and</i> ASCII.

TASK	COMMAND
Capture an interface	tcpdump -i eth0
Stop name resolution	tcpdump -i eth0 -n
Capture 100 packets	tcpdump -i eth0 -c 100
Save to a pcap file	tcpdump -i eth0 -w capture.pcap
Read a pcap file	tcpdump -r capture.pcap
Traffic to/from a host	tcpdump -i eth0 host 10.0.0.1
HTTP (port 80)	tcpdump -i eth0 tcp port 80
HTTPS / TLS (443)	tcpdump -i eth0 tcp port 443
DNS (UDP 53)	tcpdump -i eth0 udp port 53
SYN packets	tcpdump -i eth0 'tcp[13] & 2 != 0'
SYN-ACK packets	tcpdump -i eth0 'tcp[13] & 2 != 0 and tcp[13] & 16 != 0'
RST packets	tcpdump -i eth0 'tcp[13] & 4 != 0'
ICMP (ping)	tcpdump -i eth0 icmp
ARP traffic	tcpdump -i eth0 arp
Hex + ASCII view	tcpdump -i eth0 -X -n
Exclude a host	tcpdump -i eth0 not host 10.0.0.1
Verbose capture	tcpdump -i eth0 -vvv -n

TYPE / CODE	NAME
0 / 0	Echo Reply
3 / 0	Destination Network Unreachable
3 / 1	Destination Host Unreachable
3 / 2	Protocol Unreachable
3 / 3	Port Unreachable
3 / 4	Fragmentation Required
4 / 0	Source Quench
5 / 0	Network Redirect
5 / 1	Host Redirect
8 / 0	Echo Request (ping)
9 / 0	Router Advertisement
11 / 0	Time to Live Exceeded (in transit)
11 / 1	Fragment Reassembly Time Exceeded
12 / 0	Parameter Problem
13 / 0	Timestamp Request
17 / 0	Address Mask Request

BPF Filter Syntax

ELEMENT	MEANING	EXAMPLE
host	Match a host	host 10.0.0.1
src / dst	Direction qualifiers	src host 10.0.0.1, dst host 10.0.0.2
net	Match a network	net 192.168.1.0/24
port	Match a port	port 80
src port / dst port	Match source/dest port	src port 443
portrange	Match a port range	portrange 6000-6010
proto	Match a protocol	tcp, udp, icmp, ip, arp
and / &&	Both conditions	host 10.0.0.1 and tcp port 80
or /	Either condition	tcp port 80 or udp port 53
not / !	Negation	tcp and not port 22
greater / less	Match packet length	less 300
ip[0], ip[1], ...	Byte-level tests	tcp[13] & 2 != 0
src net / dst net	Source/destination network	src net
()	Grouping	(tcp or udp) and not port 53

Byte-level tests use offsets: ip[9] (protocol), tcp[13] (flags), tcp[13]&2 != 0 (SYN set).

DNS Header & Flags

FIELD	NOTES
Query ID	2 bytes — matches a query to its response.
Flags	QR, Opcode, AA, TC, RD, RA, Z, AD, CD, RCODE.
Query Count	Number of questions.
Answer Count	Number of answers.
Authority Count	Number of authority records.
Additional Count	Number of additional records.

UDP Header

FIELD	BYTES	NOTES
Source Port	2	Source service port
Destination Port	2	Destination service port
Length	2	Includes the 8-byte header; minimum value is 8
Checksum	2	Covers pseudo-header, UDP header and payload

IPv4 Header

FIELD	OFFSET	NOTES
Version	ip[0]&0xf0	4 = IPv4
Header Length (IHL)	ip[0]&0xf	In 4-byte words; minimum 5 (20 bytes)
TOS / DSCP	ip[1]	Differentiated Services
Total Length	ip[2:2]	Includes header
Identification	ip[4:2]	Unique ID for fragmented packets
Flags (DF / MF)	ip[6]	X reserved, D don't fragment, M more fragments
Fragment Offset	ip[6:2]&0x1fff	Multiply by 8
TTL	ip[8]	Number of hops before packet is discarded
Protocol	ip[9]	1 ICMP, 6 TCP, 17 UDP, 47 GRE, 50 ESP, 51 AH
Header Checksum	ip[10:2]	IP header only
Source Address	ip[12:4]	Initial sender's IP address
Destination Address	ip[16:4]	Receiving host's IP address
Options	ip[20..]	Up to 40 bytes

TCP Header

FIELD	OFFSET	NOTES
Source Port	tcp[0:2]	
Dest Port	tcp[2:2]	
Seq Number	tcp[4:4]	Increments each byte sent
Ack Number	tcp[8:4]	Next expected sequence #
Header Len	tcp[12]>>4	In 4-byte words; min 5
Reserved	tcp[12]&0xf	Set to 0
Flags	tcp[13]	CWR ECE URG ACK PSH RST SYN FIN
Window Size	tcp[14:2]	Receive window size
Checksum	tcp[16:2]	Covers pseudo-header + TCP header + payload
Urgent Pointer	tcp[18:2]	
Options	tcp[20..]	Up to 40 bytes (MSS, window scale, SACK, timestamps)

TCP Flags

SINGLE FLAGS			FLAG COMBINATIONS			SCAN PATTERNS		
FLAG	HEX	DESC	FLAGS	HEX	DESC	PATTERN	HEX	DESC
FIN	0x01	Finish	SYN-ACK	0x12	Handshake reply	NULL	0x00	No flags
SYN	0x02	Synchronize	FIN-ACK	0x11	Close + ack	SYN	0x02	Half-open scan
RST	0x04	Reset	RST-ACK	0x14	Abort + ack	FIN	0x01	FIN scan
PSH	0x08	Push	PSH-ACK	0x18	Push + ack	XMAS	0x29	FIN+PSH+URG
ACK	0x10	Acknowledgment	SYN-ACK-PSH	0x1A	Handshake + push	ACK	0x10	ACK probe
URG	0x20	Urgent	URG-ACK	0x3A	Urgent + ack	ALL	0xFF	All flags set
ECE	0x40	ECN-Echo	ECE-CWR	0xC0	ECN congestion	MAIMON	0x11	FIN+ACK scan
CWR	0x80	Congestion Reduced	ACK	0x10	Ack only	WINDOW	0x10	ACK window scan
			SYN-FIN	0x03	Inconsistent	SUPER-XMAS	0x38	XMAS + SYN+ACK
			SYN-RST	0x06	Inconsistent			
			FIN-PSH	0x09	Close + push			
			FIN-URG	0x21	Close + urgent			

Common Ports & Acronyms

TCP		UDP		ACRONYMS	
PORT	SERVICE	PORT	SERVICE	ACRONYM	MEANING
20	ftp-data	7	echo	AH	Authentication Header (RFC 2402)
21	FTP	19	chargen	ARP	Address Resolution Protocol (RFC 826)
22	SSH	53	domain (DNS)	BGP	Border Gateway Protocol (RFC 1771)
23	Telnet	67 / 68	DHCP server / client	CWR	Congestion Window Reduced (RFC 2481)
25	SMTP	69	TFTP	DF	Do Not Fragment flag (RFC 791)
43	WHOIS	123	NTP	DHCP	Dynamic Host Configuration Protocol (RFC 2131)
53	DNS	137	netbios-ns	DNS	Domain Name System (RFC 1035)
80	HTTP	138	netbios-dgm	ECN	Explicit Congestion Notification (RFC 3168)
88	Kerberos	161	SNMP	ESP	Encapsulating Security Payload (RFC 2406)
110	POP3	162	SNMP-trap	FTP	File Transfer Protocol (RFC 959)
113	authd	500	ISAKMP	GRE	Generic Route Encapsulation (RFC 2784)
119	NNTP	514	syslog	HTTP	Hypertext Transfer Protocol (RFC 1945)
143	IMAP	520	RIP	ICMP	Internet Control Message Protocol (RFC 792)
179	BGP	546 / 547	DHCPv6 client / server	IGMP	Internet Group Management Protocol (RFC 2236)
443	HTTPS	1900	SSDP	IMAP	Internet Message Access Protocol (RFC 2060)
445	MS-SMB	5353	mDNS	IP	Internet Protocol (RFC 791)
465	SMTPS			ISAKMP	Internet Security Assoc. & Key Mgmt Protocol (RFC 7296)
1433	MS-SQL			L2TP	Layer 2 Tunneling Protocol (RFC 2661)
3128	Squid			OSPF	Open Shortest Path First (RFC 1583)
3306	MySQL			POP3	Post Office Protocol v3 (RFC 1460)
3389	RDP			RFC	Request for Comments
				SMTP	Simple Mail Transfer Protocol (RFC 821)
				SSH	Secure Shell (RFC 4253)
				SSL	Secure Sockets Layer (RFC 6101)
				TCP	Transmission Control Protocol (RFC 793)
				TLS	Transport Layer Security (RFC 5246)
				TFTP	Trivial File Transfer Protocol (RFC 1350)
				TOS	Type of Service (RFC 2474)
				UDP	User Datagram Protocol (RFC 768)